V3 empirical study · Instruction

Five models · four underlyings · four volatility regimes · 1.5-year monthly calibration · return-based calibration

What this report is

- Companion to the seven-model 10,000-path V4 study. This report keeps MD-GBM, GBM, GARCH, Merton, and GARCH–Merton, whose P dynamics are estimated from lookback returns. That is the same V3 split: return-based vs option-implied Heston, not a split by the P→Q map. GBM still uses $\mu \to r_f$; GARCH uses Duan LRNVR; Merton uses Pan jump-size premium μ_J^* from listed calls on top of return-based P jumps; GARCH–Merton uses Duan LRNVR on the GARCH block plus Pan μ_J^* on jumps. MD-GBM means Markov Directional Geometric Brownian Motion: original Modified GBM with folded-normal means matched to sample $|R|$.
- Companies are SPY, AAPL, MSFT, and AMZN. No new calibration or LSM.
- Option-implied companion: V4_1p5y_monthly_empirical_study_option_implied.pdf.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 5 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Ranking column ranks 1 = best.

Experimental design (held fixed for every model)

- Models: MD-GBM (Markov Directional Geometric Brownian Motion), GBM, GARCH(1,1), Merton jump-diffusion, GARCH–Merton.
- MD-GBM means Markov Directional Geometric Brownian Motion. In this study it is specified in V4_MD-GBM_model.pdf in this folder.
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start

V3 empirical study · Instruction (continued)

Five models · four underlyings · four volatility regimes · 1.5-year monthly calibration · return-based calibration

Shared market sample and filters (not model-specific)

- For each company × regime, one listed-call sample is drawn once and reused by all five models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S, K, C; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05, valid bid–ask with relative spread ≤ 50% when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \to r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud (n_paths = 10000, seed 42, n_steps = DTE, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{model} - C_{mkt}) / C_{mkt})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE\%} = 100 \times \text{mean}(|C_{model} - C_{mkt}|/C_{mkt})$. $\text{Bias\%} = 100 \times \text{mean}((C_{model} - C_{mkt})/C_{mkt})$. Positive bias = model expensive vs market.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all five models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-3	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-3	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-3	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-3	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7–60	|S/K−1|≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end − 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date ≤ t. LSM prices the Monday contract using the latest calibration row with date ≤ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid–ask mid when the source mark is $0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All five models in this report still share that exact sample.

Heston / Heston–Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity × moneyness strata). GBM / Merton / GARCH / GARCH–Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**41.02**	**32.65**	**-4.71**	**1**
GBM	55.01	40.97	+21.16	5
GARCH	43.04	35.72	+33.41	2
Merton	51.73	37.42	+25.72	4
GARCH–Merton	44.31	37.49	+36.30	3

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**29.99**	**24.17**	**+13.83**	**1**
GBM	38.08	30.45	+24.88	5
GARCH	36.35	29.04	+22.86	2
Merton	37.32	29.83	+23.72	3
GARCH–Merton	37.90	30.58	+24.87	4

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**36.48**	**31.07**	**+14.03**	**1**
GBM	49.51	41.57	+32.90	2
GARCH	65.98	49.13	+41.26	4
Merton	60.60	52.40	+46.39	3
GARCH–Merton	68.12	51.50	+44.06	5

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**43.63**	**36.23**	**+14.85**	**1**
GBM	64.88	57.43	+44.13	4
GARCH	53.62	42.08	+35.24	3
Merton	70.92	64.09	+53.08	5
GARCH–Merton	52.76	41.90	+35.13	2

RMSE% is 100×√mean(((model−market)/market)²). MAE% is 100×mean(|model−market|/market). Bias% is 100×mean((model−market)/market). Bias% > 0 means the model is expensive vs th

Page 5 · Table block · Company summary · five models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookback, rolling=monthly, n_paths=

Table Co-1 Company summary · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · n̄ = 50

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	37.97	29.51	+5.30	2
GBM	49.54	37.95	+23.71	5
GARCH	**37.65**	**31.10**	**+23.97**	**1**
Merton	47.89	36.89	+30.25	4
GARCH–Merton	45.49	38.29	+35.63	3

Table Co-2 Company summary · 2013-2014 Normal
2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · n̄ = 50

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**30.15**	**24.58**	**+14.84**	**1**
GBM	47.93	39.44	+36.05	3
GARCH	53.53	43.66	+39.14	4
Merton	46.87	37.36	+34.12	2
GARCH–Merton	75.13	64.41	+64.13	5

Table Co-3 Company summary · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · n̄ = 51

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**27.95**	**24.23**	**+7.20**	**1**
GBM	39.31	33.51	+23.84	3
GARCH	36.84	29.16	+21.78	2
Merton	43.10	37.05	+32.25	4
GARCH–Merton	49.41	40.98	+37.98	5

Table Co-4 Company summary · 2019-2020 COVID
2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · n̄ = 50

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**34.58**	**27.59**	**+9.56**	**1**
GBM	48.18	39.74	+30.10	3
GARCH	41.61	31.18	+23.29	2
Merton	52.19	44.42	+37.76	4
GARCH–Merton	52.64	41.77	+36.69	5

RMSE%, MAE%, and Bias% are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMSE% only.

Page 6 · Table block · AAPL · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	43.89	32.72	+12.84	3
GBM	57.67	42.83	+31.10	5
GARCH	**34.36**	**28.77**	**+25.46**	**1**
Merton	55.37	40.83	+36.40	4
GARCH–Merton	40.33	34.50	+32.14	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**30.36**	**23.26**	**+17.13**	**1**
GBM	50.20	39.70	+38.63	4
GARCH	40.08	29.28	+23.86	2
Merton	48.87	38.01	+37.02	3
GARCH–Merton	59.86	50.37	+50.37	5

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**23.64**	**20.27**	**+5.37**	**1**
GBM	32.53	28.14	+18.76	2
GARCH	40.82	29.83	+21.75	4
Merton	36.65	31.82	+28.29	3
GARCH–Merton	54.71	44.78	+41.77	5

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**31.97**	**25.38**	**+9.95**	**1**
GBM	46.19	39.52	+30.76	3
GARCH	45.53	33.84	+29.35	2
Merton	47.03	40.14	+35.87	4
GARCH–Merton	57.26	46.26	+44.06	5

RMSE% is $100\times\sqrt{\text{mean}(((\text{model}-\text{market})/\text{market})^2)}$. MAE% is $100\times\text{mean}(|\text{model}-\text{market}|/\text{market})$. Bias% is $100\times\text{mean}((\text{model}-\text{market})/\text{market})$. Bias% > 0 means the model is expensive vs th

Page 7 · Table block · MSFT · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**34.43**	**27.73**	**+3.32**	**1**
GBM	44.02	34.76	+17.99	3
GARCH	50.93	41.05	+35.87	4
Merton	41.80	34.10	+23.08	2
GARCH–Merton	61.38	51.11	+48.59	5

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**38.79**	**31.25**	**+28.02**	**1**
GBM	62.49	53.14	+52.98	3
GARCH	85.94	73.78	+73.69	4
Merton	56.13	47.20	+46.76	2
GARCH–Merton	108.47	97.36	+97.36	5

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**25.61**	**21.67**	**+4.03**	**1**
GBM	41.73	34.38	+27.55	3
GARCH	37.25	30.23	+24.44	2
Merton	43.31	35.57	+33.24	4
GARCH–Merton	48.79	40.81	+38.95	5

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**31.75**	**23.91**	**+4.32**	**1**
GBM	45.93	37.46	+28.12	2
GARCH	49.20	34.48	+25.36	3
Merton	50.06	41.56	+34.80	4
GARCH–Merton	54.17	40.27	+34.17	5

RMSE% is 100×√mean(((model−market)/market)²). MAE% is 100×mean(|model−market|/market). Bias% is 100×mean((model−market)/market). Bias% > 0 means the model is expensive vs th

Page 8 · Table block · AMZN · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	35.59	28.09	-0.26	3
GBM	46.92	36.28	+22.05	5
GARCH	**27.67**	**23.47**	**+10.57**	**1**
Merton	46.51	35.73	+31.26	4
GARCH–Merton	34.76	29.27	+26.15	2

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	**21.30**	**19.23**	**-0.63**	**1**
GBM	31.09	25.49	+16.54	2
GARCH	34.57	27.93	+19.87	3
Merton	35.62	26.86	+18.58	4
GARCH–Merton	57.06	45.50	+44.68	5

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	34.59	30.76	+12.20	2
GBM	43.66	38.02	+25.21	3
GARCH	**32.44**	**27.42**	**+19.15**	**1**
Merton	49.34	43.76	+35.22	5
GARCH–Merton	44.74	37.34	+33.22	4

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	40.03	33.48	+14.41	2
GBM	52.42	42.24	+31.41	4
GARCH	**30.12**	**25.22**	**+15.17**	**1**
Merton	59.46	51.55	+42.61	5
GARCH–Merton	46.47	38.78	+31.83	3

RMSE% is 100×√mean(((model−market)/market)²). MAE% is 100×mean(|model−market|/market). Bias% is 100×mean((model−market)/market). Bias% > 0 means the model is expensive vs th

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells bes
1	**MD-GBM**	**33.94**	**34.51**	**12**
2	GARCH	44.24	40.45	4
3	GBM	47.65	46.56	0
4	Merton	49.42	49.10	0
5	GARCH–Merton	54.44	53.47	0

Table S2 · Wins by company (lowest RMSE% in that company×regi

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
MD-GBM	**4**	**3**	**4**	**1**	**12**
GBM	0	0	0	0	0
GARCH	0	1	0	3	4
Merton	0	0	0	0	0
GARCH–Merton	0	0	0	0	0

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
MD-GBM	**38.73**	**30.11**	**30.08**	**36.85**	**33.94**
GBM	50.90	45.47	41.86	52.35	47.65
GARCH	39.00	49.23	44.12	44.61	44.24
Merton	48.85	44.48	47.47	56.87	49.42
GARCH–Merton	45.20	65.82	54.09	52.67	54.44

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Page 8 · Summary by company · mean RMSE% across four regimes

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
MD-GBM	**37.78**	**38.75**	**4**
GBM	51.87	52.26	0
GARCH	49.75	48.33	0
Merton	55.14	56.16	0
GARCH–Merton	50.77	48.54	0

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
MD-GBM	**32.47**	**31.17**	**3**
GBM	46.65	48.20	0
GARCH	40.20	40.45	1
Merton	46.98	47.95	0
GARCH–Merton	53.04	55.99	0

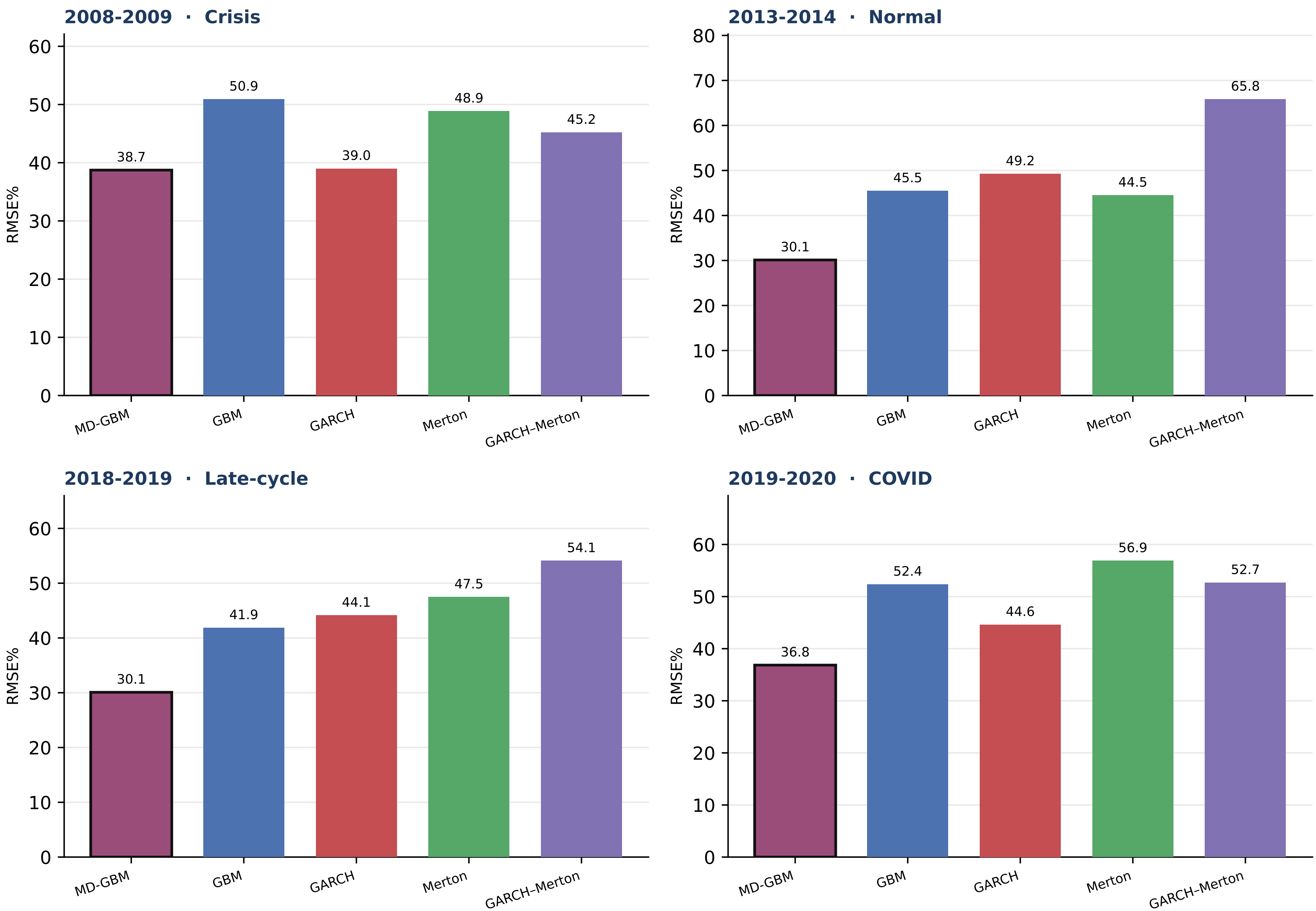
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
MD-GBM	**32.65**	**33.09**	**4**
GBM	48.54	44.98	0
GARCH	55.83	50.06	0
Merton	47.83	46.69	0
GARCH–Merton	68.20	57.78	0

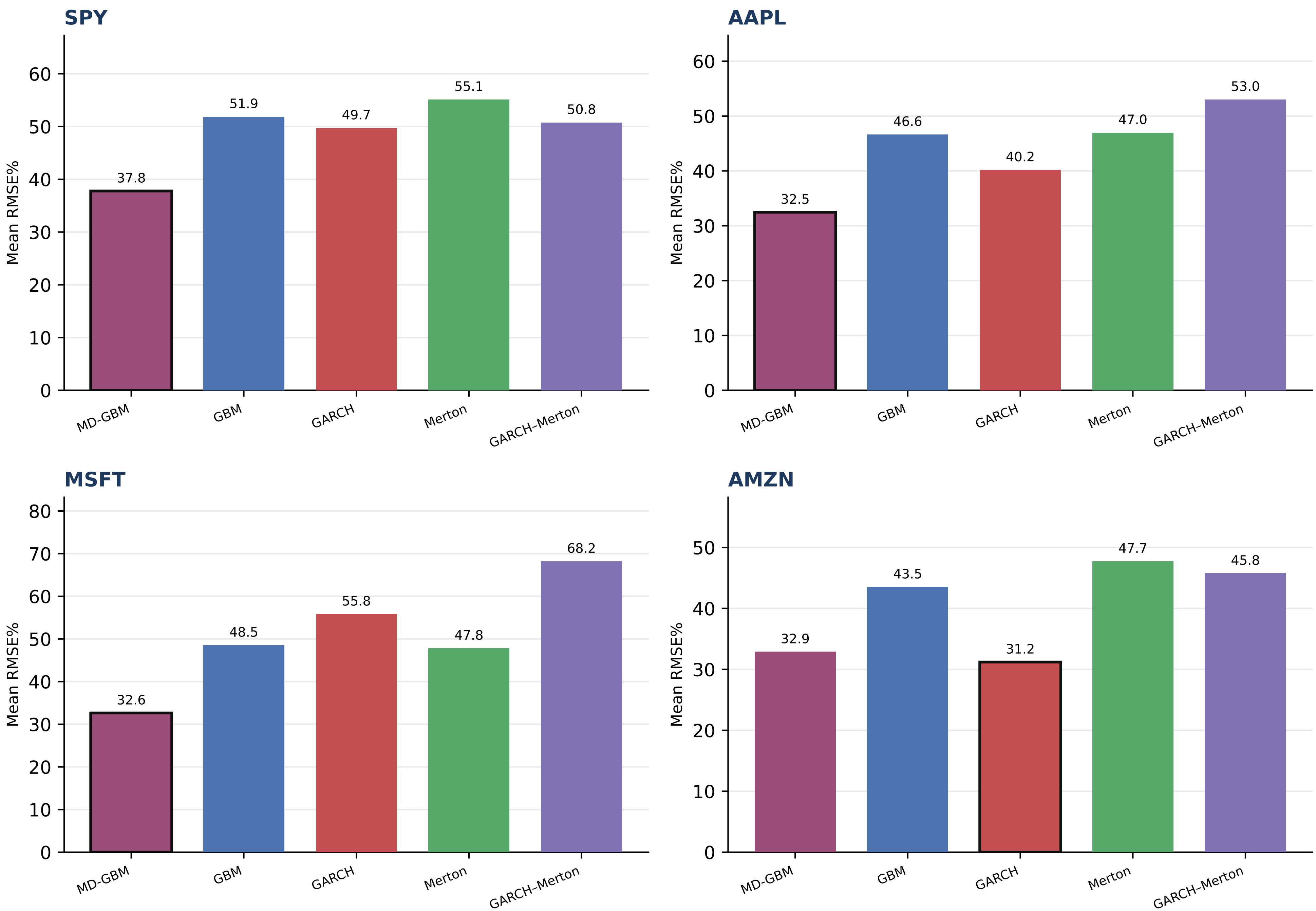
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
MD-GBM	32.88	35.09	1
GBM	43.52	45.29	0
GARCH	**31.20**	**31.28**	**3**
Merton	47.73	47.93	0
GARCH–Merton	45.76	45.60	0

Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.

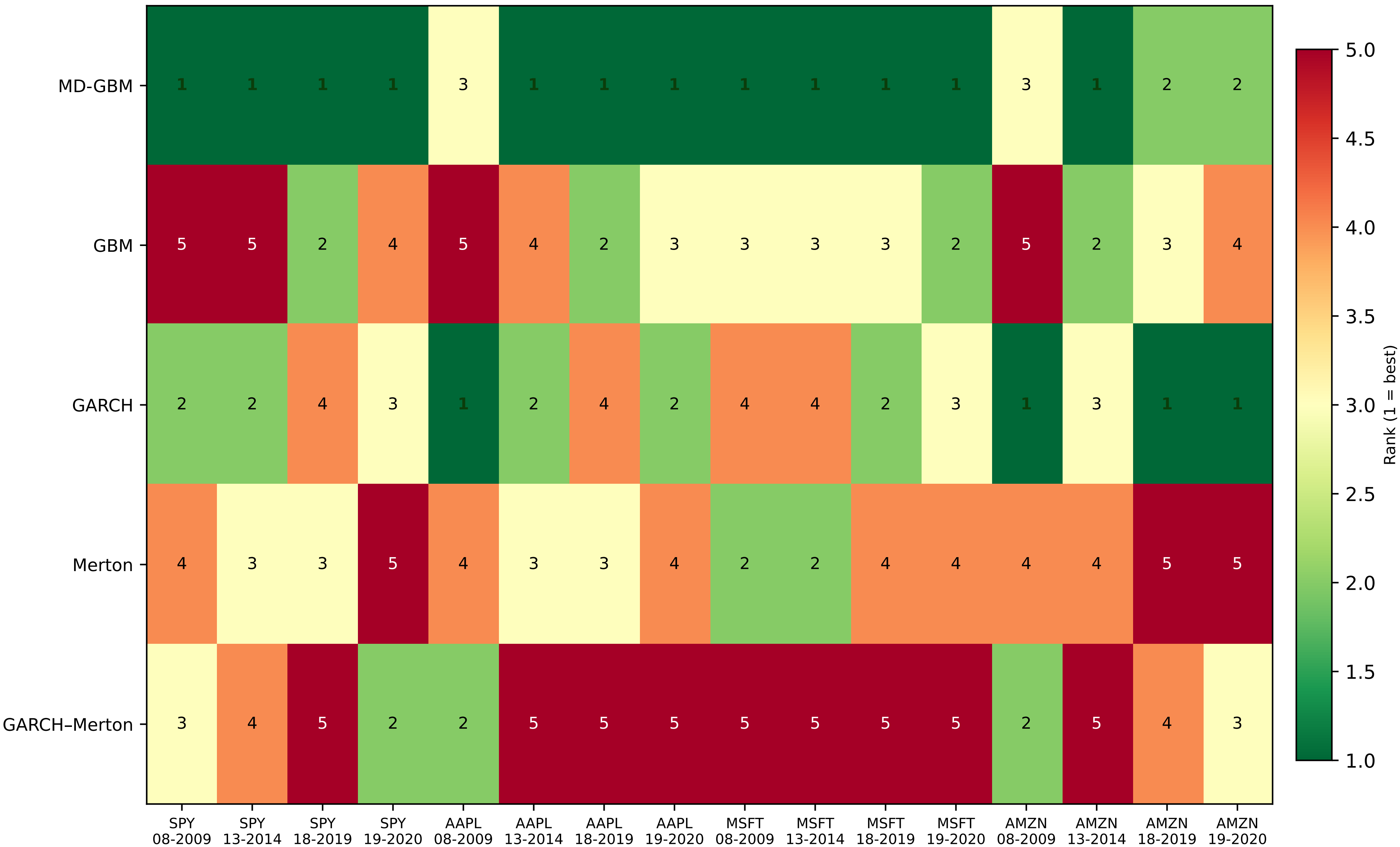


Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.

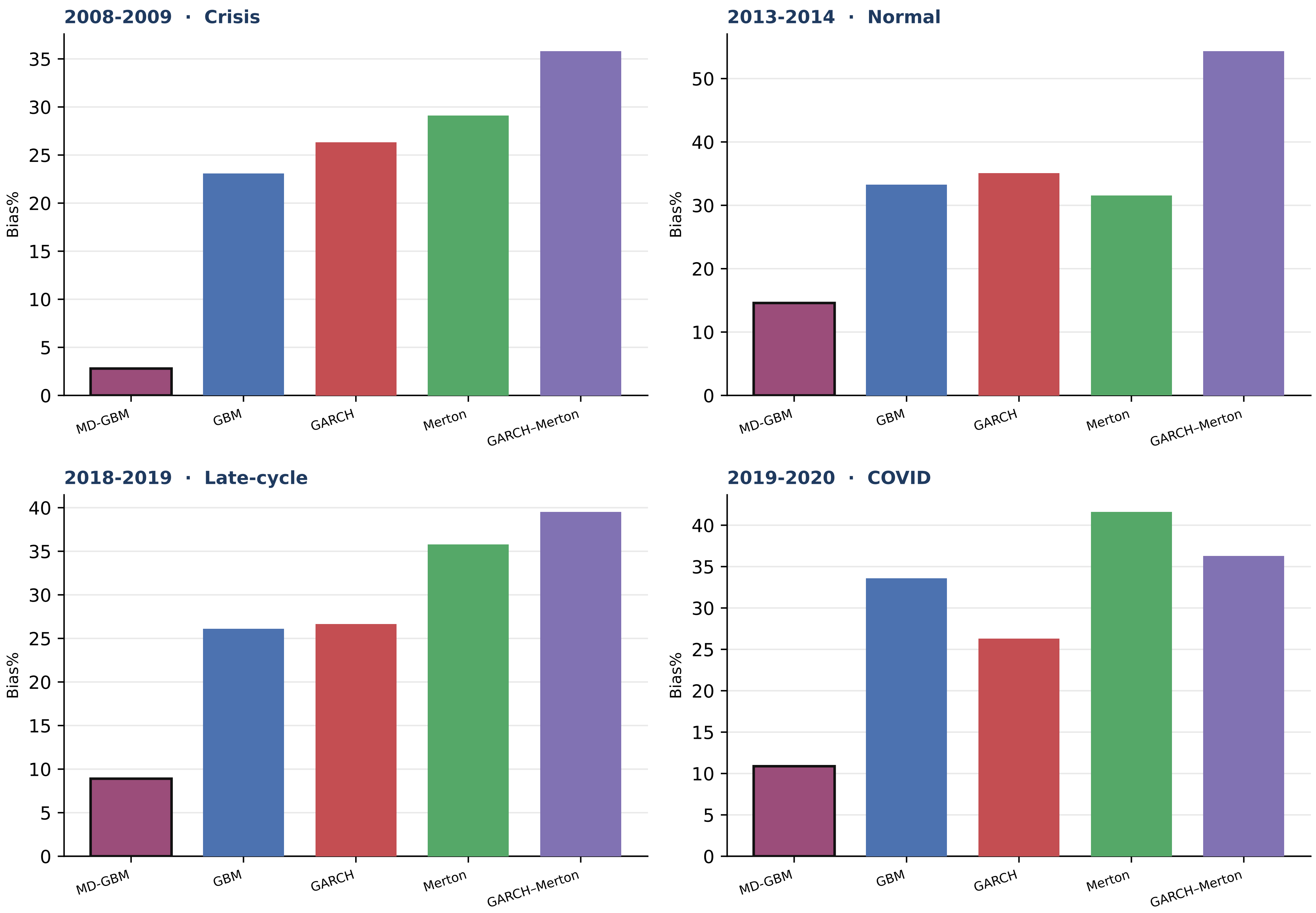


Outlined bar = lowest mean RMSE% for that company.

Page 11 · Figure 3 · Model rank by company × regime (1 = best RMSE%)



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero Bias% (not the RMSE ranking). Positive = model expensive vs listed quote.

Page 13 · Final analysis

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, MD-GBM is the best overall model (mean RMSE% = 33.94; 12 of 16 cells).
- Second overall is GARCH (mean RMSE% = 44.24; 4 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: GARCH, MD-GBM.
- SPY: lowest mean RMSE% is MD-GBM (37.78); wins 4 of 4 regimes.
- AAPL: lowest mean RMSE% is MD-GBM (32.47); wins 3 of 4 regimes.
- MSFT: lowest mean RMSE% is MD-GBM (32.65); wins 4 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (31.20); wins 3 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is MD-GBM (38.73).
- 2013-2014 (Normal): best mean RMSE% is MD-GBM (30.11).
- 2018-2019 (Late-cycle): best mean RMSE% is MD-GBM (30.08).
- 2019-2020 (COVID): best mean RMSE% is MD-GBM (36.85).
- No-jump models as a group have lower mean RMSE% (41.94) than the jump models in this report (51.93).
- For the overall winner (MD-GBM), crisis RMSE% is 38.73 vs 30.11 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1…MSFT-4 uses exactly the same option contracts for all five models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE% and Bias% are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).